BEFORE YOU READ

The identities in six groups

The algebraic identities on this page group by the kind of expression they expand or factor.

- L0 **Algebraic identities**
  - L1 **Squares of binomials**
    - L2 **Square of a sum** $(a+b)^2$
    - L2 **Square of a difference** $(a-b)^2$
    - L2 **Difference of squares** a^2-b^2
  - L1 **Cubes of binomials** $(a\pm b)^3$
  - L1 **Higher powers** the binomial theorem $(a+b)^n$
  - L1 **Trinomial square** $(a+b+c)^2$
  - L1 **Sums and differences of powers**
    - L2 **Sum and difference of cubes** $a^3\pm b^3$
    - L2 **Difference of nth powers** a^n-b^n
    - L2 **Sum of nth powers** factors by a + b when n is odd a^n+b^n
  - L1 **Useful specials**
    - L2 **Sophie Germain** a^4+4b^4
    - L2 **Sum of squares over $\mathbb{C}$** a^2+b^2

TAKEAWAY

Read them both ways

Every identity works left to right for expanding and right to left for factoring.